

Tilt switch

DETECT

INPUT

What does it do?

A tilt switch changes state when it tips past a particular angle.

A tiny metal ball rolls inside and touches two contacts, completing the circuit.

It can detect that an object has been picked up, tipped over or opened.

Uses



Iron safety shut-off

DETECT
Iron tips
sideways



DECIDE
Shut-off
delay
reached



DO
Switch the
heater off



Grumpy pencil case

DETECT
Case is tipped



DECIDE
Pencils are
escaping



DO
Make a
grumpy noise



Metal can
Contains the
rolling ball

Legs
No polarity

Angle
Changes the
contact

Code

● Raspberry Pi

```
from gpiozero import Button  
  
tilt = Button(23)  
  
tilted = tilt.is_pressed
```

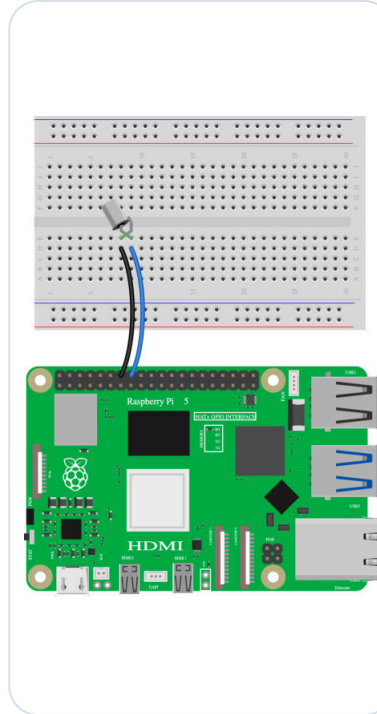
● Raspberry Pi Pico

```
from picozero import Switch  
  
tilt = Switch(16)  
  
tilted = tilt.is_closed
```

The reading may flicker while the ball rolls.
Check it more than once before treating it as a definite tilt.

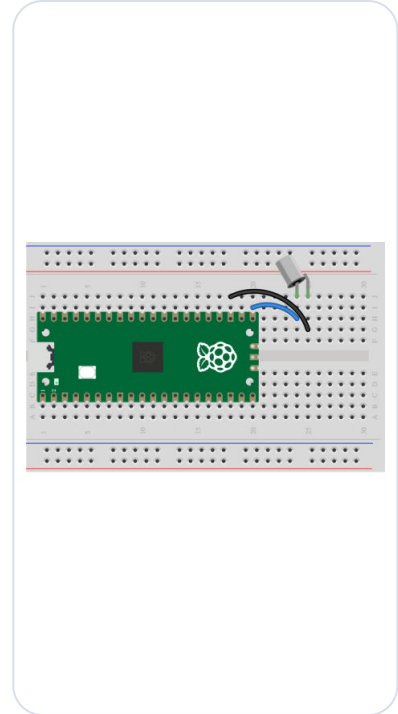
Wiring

RASPBERRY PI 5



GPIO 23 → tilt switch → GND

PICO



GP16 → tilt switch → GND